

RuralGrow AI

Himalayan Micro-Merchant Growth & AI Assistant

TBI-GEU Summer Internship Final Presentation Deliverable

Intern Details

- Name: **Saurabh Parihar**
- Intern ID: **TBI-26100640**
- University: **Graphic Era University (GEU)**
- Domain: **Full-Stack Web Dev & AI / DevOps**

Public Verification Links

- Web App (Vercel):
<https://rural-grow-58fsd5yhj-rural-grow-ai.vercel.app>
- REST API (Render):
<https://ruralgrow-ai.onrender.com>
- GitHub Repository:
<https://github.com/Saurabhparihar12/RuralGrow-AI>

2. Problem Statement & Mission

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Ø=P̈ The Core Problem

- **Language & Literacy Barrier:**
Rural merchants (farmers, weavers, homestay hosts) struggle to write for
- **Zero Marketing Expertise:**
High quality local products lack digital visibility on Instagram, Facebook, a
- **Review Ignorance:**
Unanswered customer reviews lead to lost sales and negative online repu
- **Lack of Business Guidance:**
Limited access to experts regarding PM-KISAN, crop pricing, and digital e

Ø=Ü; The RuralGrow AI Solution

- **AI Review Sentiment & Auto-Reply:**
Instantly analyzes customer reviews and drafts polite responses in regional Indian languages.
- **Instant Social Copywriter:**
Generates promotional posts with custom discounts tailored for WhatsApp, Instagram, and FB.
- **HimalayaGrow AI Assistant:**
24/7 conversational guide for scheme updates, pricing, and agricultural advice.
- **Visual Growth Dashboard:**
Interactive analytics tracking review ratings and customer feedback metrics.

3. Technology Stack & Architectural Rationale

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Frontend: React 19 + Vite + Tailwind CSS

Rationale: Lightning-fast SPA rendering, component reusability, and modern responsive UI styling.

Backend: Node.js + Express.js REST API

Rationale: Non-blocking I/O event loop, seamless JSON handling, and modular controller architecture.

Database: MongoDB Atlas (Mongoose ORM)

Rationale: Flexible document schema for nested reviews/captions, high scalability, and cloud persistence.

AI Integration: Google Gemini 1.5 Flash API

Rationale: High-speed multi-lingual Indian language generation with zero cost on developer free tier.

Auth & Security: JWT + Passport Google OAuth

Rationale: Stateless session security, BCrypt hashing, CORS restriction, and Helmet security headers.

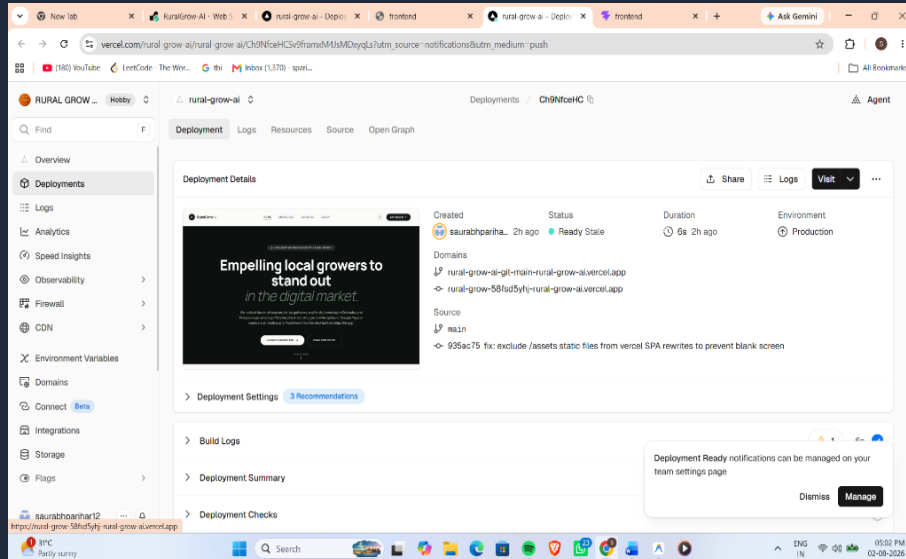
Deployment: Vercel (Client) + Render (API)

Rationale: Automated CI/CD from GitHub main branch with zero server management overhead.

4. Frontend User Experience & Live Interface

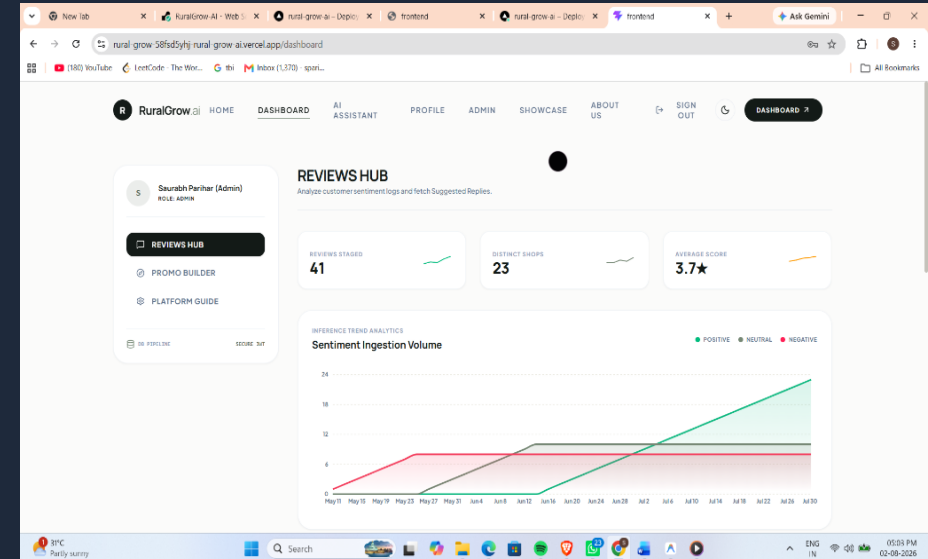
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Live Vercel Application Homepage



- Responsive Glassmorphism design with Himalayan dark/light theme.
- Live URL: <https://rural-grow-58fsd5yhj-rural-grow-ai.vercel.app>

Merchant Growth & Review Analytics Dashboard



- Recharts sentiment breakdown (Positive / Neutral / Negative).
- Direct action buttons to reply or generate marketing posts.

5. Core Backend REST APIs & Verification

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API 1: POST /api/ai/analyze-review

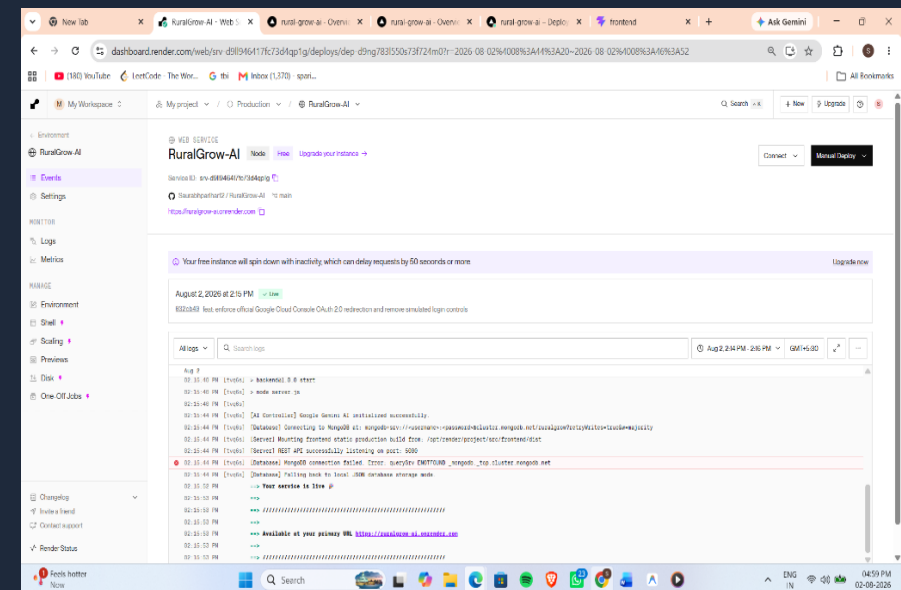
- Working:
Accepts customer review text & rating -> analyzes sentiment -> returns suggested polite response in Hindi/English.
- Resiliency:
Equipped with local AI simulation fallback if Gemini API is unreachable.

[API 1 Test Payload & Response]

```
REQUEST: { "reviewText": "Awesome organic honey!", "rating": 5 }  
RESPONSE: {  
  "success": true,  
  "sentiment": "Positive",  
  "reply": "Dhanyawaad! We are glad you enjoyed our raw honey."  
}
```

API 2: POST /api/ai/chat

- Working:
Accepts merchant query -> runs prompt engineering -> returns advice tailored for Himalayan micro-merchants.
- Health Verification:
Render API Endpoint: GET /api/health -> 200 OK.



6. Database Choice & Schema Architecture

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Database Selection: MongoDB Atlas

- Why MongoDB Atlas?

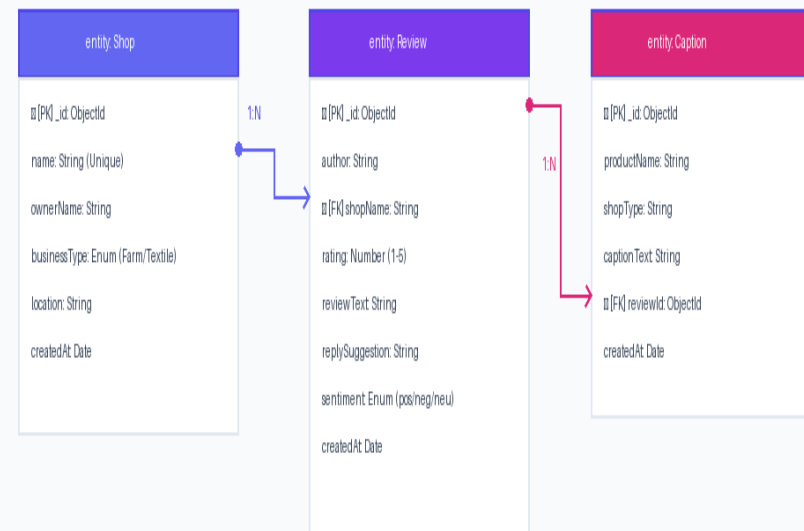
1. Flexible JSON document structure fits reviews, captions & profiles.
2. Cloud Atlas cluster ensures zero local setup overhead.
3. Fast Mongoose indexing on User and Review IDs.

- Models Designed:

- User Schema (Auth, Role, ShopName)
- Review Schema (Sentiment, Rating, AI Reply)
- Caption Schema (Platform, Discount, Content)
- Shop Schema (Location, Category, Contact)

Entity Relationship & Schema Diagram

RURALGROW AI - CORE DATABASE SCHEMA RELATIONSHIP MODEL



Production Stack: MongoDB Atlas Persistent Database, Express.js REST Engine, Mongoose ODM

7. AI Features & Gemini LLM Integration

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LLM Model: Google Gemini 1.5 Flash

- Primary Use Cases:

1. Review Re-Composer: Translates customer feedback into professional replies.
2. Social Post Writer: Formats promo text for WhatsApp/FB with emojis.
3. HimalayaGrow Advisory: Answers PM-KISAN & pricing questions.

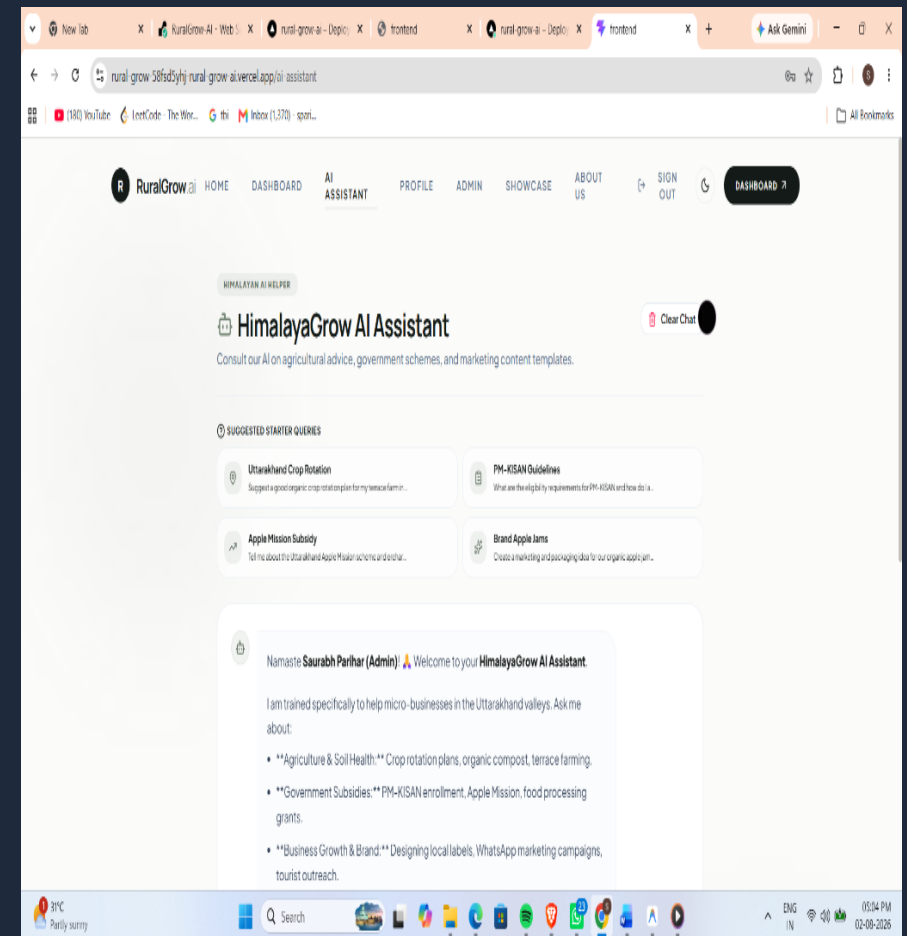
- System Prompting:

Configured with specialized system instructions to maintain a polite, helpful Himalayan assistant persona.

- Resilient Fallback:

Offline rule engine ensures 100% uptime if API key is rate limited.

Live HimalayaGrow AI Chat Assistant



8. Production Hosting & Cloud Infrastructure

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%² Vercel Cloud (Frontend)

- Platform:
Vercel Edge Network for React SPA.
- Configuration:
vercel.json rewrite rules for SPA client routing.
- Deployment URL:
<https://rural-grow-58fsd5yhj-rural-grow-ai.vercel.app>
- Key Benefits:
Instant global CDN distribution, SSL HTTPS by default, and continuous deployment from GitHub main branch.

&™ Render Web Service (Backend)

- Platform:
Render Cloud Web Service for Node.js Express API.
- Configuration:
Infrastructure-as-code render.yaml manifest.
- Deployment URL:
<https://ruralgrow-ai.onrender.com>
- Key Benefits:
Automated Node build environment, dynamic port binding, zero secret exposure, and health check monitoring.

9. Public Links & Verification Matrix

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✓ Verified Publicly Accessible Links

All links below are live, public, and accessible without login barriers:

Live Frontend Web App (Vercel)

<https://rural-grow-58fsd5yhj-rural-grow-ai.vercel.app>

ACTIVE ✓

Live Backend REST API (Render)

<https://ruralgrow-ai.onrender.com>

ACTIVE ✓

Backend API Health Check Endpoint

<https://ruralgrow-ai.onrender.com/api/health>

200 OK ✓

GitHub Public Source Repository

<https://github.com/Saurabhparihar12/RuralGrow-AI>

VERIFIED ✓

Official Submission ZIP Deliverable

W9_Submission_TBI-26100640.zip in root directory

VERIFIED ✓

10. Internship Reflection & Key Learnings

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Ø>Ýà Technical & DevOps Skills Learned

1. Full-Stack MERN Architecture:
Mastered building modular React frontend components with Node.js Express REST API integration.
2. Production Cloud Deployment:
Configured Vercel rewrites, Render web services, dynamic CORS security, and environment variables.
3. Practical LLM Engineering:
Implemented prompt engineering with Google Gemini API and built offline fallback simulators.
4. Database Modeling:
Designed relational document schemas in MongoDB Atlas for persistent storage.

Ø<ß Overall Internship Experience

- Real-World Impact:
Building RuralGrow AI provided deep insights into solving grassroots problems for Himalayan micro-merchants.
- Guidance & Mentorship:
Grateful to Mayank Kumar sir and the Skill Dev Team at TBI-GEU for structured weekly milestones and feedback.
- Industry Standard Habits:
Adopted git workflows, clean documentation standards, API contract design, and automated PDF reporting.